



Newson's \$12,000 hourglass

Apple's designer Marc Newson is selling an expensive hourglass in collaboration with Hodinkee. <http://dgit.in/MarcHourglass>



Find My Device on Gear S3

Samsung's latest update for its Gear S3 smartwatches finally includes Google's Find My Device. <http://dgit.in/GearS3fmd>



Abiogenesis

The fascinating story of how life first emerged in a non-living world.

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How big is the universe?
What really is time?
How did life begin on earth?

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he quest for finding these answers has kept us occupied ever since we first started thinking about our

place in the cosmos. For the longest time, the standard explanation for everything was, "...the Gods did it", and dissenters were routinely branded heretics, tortured, even killed.

Fortunately for us, times change. Over the last few centuries, we have devised wildly ambitious experiments to unlock the secrets surrounding these mysteries. As for life, they've even tried to recreate it in the lab, and while no one has been successful – we've found some fascinating answers.

This is the story of abiogenesis, the process of how life first originated from, well, not life. In order to answer how life began, we must first look at where and when it began. Buckle up because our first stop takes us 4.6 billion years ago.

EARTH BEFORE LIFE

The Earth always was blue green globe with oceans, clouds, terrain, teeming with plant and animal life. The Hadean Earth (period lasting from 4.6 billion to 4 billion years ago) was marked by its inhospitable environment made of a

thick blanket of carbon dioxide, carbon monoxide, hydrogen sulfide, methane, ammonia, and other chemically reducing gases, constant lightning strikes, surface temperatures of around 230 degrees Celsius. With hardly any surface to speak of, it was constantly bombarded by meteorite showers into a hot molten pulp, and the atmospheric pressure was so high that water remained in the liquid state even at temperatures well above its boiling point.

The first person who attempted to study the preconditions that led to the creation of life was a Soviet scientist named Alexander Oparin. In 1924, he published *The Origins of Life* and laid out the model for how Earth cooled down following the Hadean period, creating the conditions for the emergence of sugars and amino acids in the Earth's waters – molecules central to life.



Is it me or does early Earth pretty much look like the biblical description of hell?

Credit: Tim Bertelink